STATISTICS

Every Statistics Concept a Data Scientist Needs at ₹20-30 LPA

The complete list of 24 statistics concepts tested in mid-to-senior data scientist interviews, grouped into 6 areas, each explained in plain words.

INSIDE THIS PDF · 9 PAGES

- **The 6 areas, 24 concepts**
- **1. Descriptive statistics**
- **2. Probability**
- **3. Inference**
- **4. Experimentation**
- **5. Regression**
- **6. Causal thinking**

@thedataguy16

The 6 areas, 24 concepts

At this level you are expected to choose the right method and explain its limits, not just define terms.

- **1. Descriptive statistics**
  4 concepts
- **2. Probability**
  4 concepts
- **3. Inference**
  4 concepts
- **4. Experimentation**
  4 concepts
- **5. Regression**
  4 concepts
- **6. Causal thinking**
  4 concepts

Score yourself on the last page. 20+ means you are ready for this band.

@thedataguy16

1. Descriptive statistics

Summarising data correctly before modelling anything.

Mean vs median

Median for skewed data like income or order value. Mean is pulled by outliers.

Variance and standard deviation

How spread out values are. SD is in the same units as the data.

Percentiles and IQR

p50, p90, p99 for latency or delivery time. IQR = Q3 − Q1, used for outlier rules.

Skewness

Long right tail is common in business data. Log-transform before many models.

@thedataguy16

2. Probability

The base of every model and test.

- **Conditional probability and Bayes**

  P(A|B) = P(B|A) × P(A) / P(B). Classic: positive test result for a rare disease is still likely a false positive.

- **Normal distribution**

  Heights, measurement error, sample means. 68-95-99.7 rule.

- **Binomial and Poisson**

  Binomial: number of conversions in n visits. Poisson: number of events per hour.

- **Expected value**

  Average outcome weighted by probability. Used for pricing, risk and decision questions.

@thedataguy16

3. Inference

Going from a sample to a conclusion about everyone.

- **Central Limit Theorem**

  Means of large samples are roughly normal, whatever the data looks like. Why t-tests work on skewed data.

- **Confidence interval**

  A range that would contain the true value in 95% of repeated samples. Report it with every estimate.

- **p-value**

  If the null were true, the chance of a result this extreme. It is not the chance the null is true.

- **Type I, Type II and power**

  False positive, false negative, and the chance of detecting a real effect (aim for 80%).

@thedataguy16

4. Experimentation

The most tested area at product companies.

Sample size

Set by baseline rate, minimum detectable effect, significance and power. Decide before starting.

Peeking problem

Checking daily and stopping when significant inflates false positives. Fix the duration or use sequential methods.

Multiple testing

20 metrics at 0.05 gives about 1 false win by chance. Use one primary metric or Bonferroni / FDR correction.

Novelty and network effects

Early lift fades. Users influence each other. Run long enough and consider cluster randomisation.

@thedataguy16

5. Regression

Interviewers ask about assumptions and interpretation, not just fitting.

- **Linear regression assumptions**

  Linear relationship, independent errors, constant variance, no strong multicollinearity.

- **Coefficient interpretation**

  "Holding other features constant, each extra ₹1,000 in income adds X to spend."

- **Logistic regression and odds**

  Coefficients are log-odds. exp(coef) is the odds ratio.

- **Multicollinearity**

  Correlated features make coefficients unstable. Check VIF, drop or combine features.

@thedataguy16

6. Causal thinking

What separates ₹30 LPA candidates: knowing when data cannot prove cause.

- **Correlation vs causation**

  Ice cream sales and drownings both rise in summer. Heat drives both.

- **Confounders**

  A third variable affecting both cause and effect. Control for it or randomise.

- **Selection bias**

  Analysing only users who stayed hides why others left.

- **Difference-in-differences**

  When you cannot run an A/B test: compare the change in a treated group to the change in a similar untreated group.

@thedataguy16

Score yourself

Count the concepts you can explain with a business example, without notes.

- **20-24**

  Ready for ₹20-30 LPA data scientist interviews on the statistics side.

- **14-19**

  Focus on Experimentation and Causal thinking. They are the most tested at this level.

- **Below 14**

  Start with Inference. Everything in Experimentation depends on it.

Interview tip: for every concept, have one example from your own work or projects. Definitions alone sound junior.

@thedataguy16